Holographic RIS-Aided Reference Energy Detection for Low-Complexity Noncoherent Communications

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***Abstract*—This paper presents a comprehensive framework for signal recovery in holographic reconfigurable intelligent surface (RIS) receivers, where only amplitude information is available at the receiver. The proposed system employs a known reference wave to enable phase retrieval from intensity-only measurements. We formulate the complete signal model from transmission to reception, and propose a Gerchberg-Saxton algorithm to estimate the wave vector and complex channel gain. Building upon this, we develop a Gauss-Newton algorithm for efficient signal recovery to conduct real-time estimation of the symbols utilizing the property of rapid convergence. Simulation results validate the effectiveness of the proposed approach, showing accurate signal recovery under various signal-to-noise ratio conditions. The complete modeling framework and algorithm implementation details are provided to facilitate reproducibility and further research in RIS-based communication systems.**

***Index Terms*—Reconfigurable Intelligent Surface, holographic receiver, phase retrieval, signal recovery, amplitude-only measurements.**

I. Introduction

Reconfigurable intelligent surfaces (RISs) have emerged as a promising technology for next-generation wireless communication systems, owing to their ability to reshape the wireless propagation environment in a programmable and energy-efficient manner [1], [2]. By adaptively controlling the electromagnetic response of a large number of low-cost reflecting elements, RISs can provide additional degrees of freedom for coverage enhancement, interference management, and spatial signal manipulation. More recently, the functionality of RISs has been extended beyond passive beamforming toward active reception and signal processing, giving rise to the concept of holographic RIS receivers [3]. By exploiting densely deployed metasurface elements, holographic RIS receivers are expected to capture rich spatial information of impinging electromagnetic waves while maintaining much lower hardware complexity than conventional antenna arrays equipped with fully digital radio-frequency chains.

Despite these advantages, a fundamental bottleneck of holographic RIS receivers lies in the acquisition of complex-valued signal information. Conventional coherent receivers rely on down-conversion, local oscillator synchronization, and in-phase/quadrature sampling to obtain both amplitude and phase information. In contrast, practical RIS-based or metasurface-based receivers may only have access to intensity or power measurements due to hardware, power, and integration constraints [4]. As a result, the received signal phase is not directly observable, and the recovery of complex-valued symbols from intensity-only measurements naturally leads to a phase retrieval problem, which has been extensively studied in optics and signal processing [5], [6].

A promising way to alleviate this ambiguity is to introduce a known reference wave before intensity detection. Inspired by optical holography, where a reference beam interferes with an object wave to encode both amplitude and phase information into an intensity pattern [7], a reference-assisted RIS receiver superimposes a known radio-frequency reference signal with the unknown received signal at the metasurface or receiver front end. The resulting square-law measurement contains not only the power of the unknown signal but also a reference-induced interference term. This interference term serves as a phase anchor and makes it possible to infer the hidden complex-valued signal from real-valued intensity observations.

[8] investigated the application of phase retrieval techniques for channel estimation in orthogonal frequency-division multiplexing (OFDM) systems, demonstrating that accurate channel estimation is possible even when only magnitude information is available, which shows the phase retrieval problems often appear in the communication area. [9] proposed an affine phase retrieval framework for wireless communication systems, where the received signal is modeled as an affine transformation of the transmitted signal, and developed algorithms for signal recovery under this model, where the problems formulation is very similar to the reference-assisted phase retrieval model in this paper.

However, existing studies on RIS-assisted reception and

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phase retrieval still leave several key questions insufficiently addressed. Most RIS receiver designs focus on enhancing the received signal power, improving beamforming gain, or reducing hardware complexity, while the intrinsic identifiability of complex symbols under intensity-only measurements has not been fully characterized. In particular, when the receiver observes only the squared magnitude of the superimposed field, different transmitted symbols may produce identical or nearly identical intensity patterns. In such cases, no detector, including maximum-likelihood detection or iterative phase retrieval algorithms, can reliably distinguish the transmitted symbols. Therefore, the central issue is not only how to design an efficient recovery algorithm, but also whether the transmitted symbols are uniquely encoded in the intensity domain in the first place.

This issue becomes more critical in holographic RIS receivers because the measured intensity pattern is jointly determined by multiple coupled components, including the reference wave, the pulse-shaping or sampling operation, and the RIS array manifold. If the effective matrix maps two different symbol vectors to the same field response, the corresponding intensity measurements remain indistinguishable regardless of how the detector is implemented. Likewise, even when the effective field responses are different, an improperly designed reference wave may fail to project their phase difference into the observable intensity domain.

Consequently, the lack of a systematic identifiability analysis limits the theoretical understanding and practical design of reference-assisted holographic RIS receivers. Existing phase retrieval methods often assume generic random measurements or focus on continuous signal recovery, whereas communication symbols belong to finite constellations with specific algebraic structures. This discrete nature creates both opportunities and challenges: on one hand, exact recovery may be possible under weaker conditions than those required for general continuous phase retrieval; on the other hand, constellation-dependent ambiguities may still arise if the reference wave and the RIS measurement structure are not properly designed. Therefore, it is essential to investigate how the reference wave, the shaping matrix, and the RIS array manifold jointly determine the uniqueness and stability of symbol recovery.

In this work, we address this problem by developing a reference-assisted phase retrieval framework for holographic RIS receivers. Instead of treating the reference signal merely as an auxiliary hardware component, we characterize its role as a phase-anchoring mechanism that converts hidden phase information into measurable intensity variations. We analyze the discrete identifiability condition of transmitted symbols under the reference-assisted intensity observation model and show that successful recovery requires two complementary properties: the effective RIS measurement matrix must preserve the differences between constellation points, and the reference wave must separate these differences in the intensity domain. Based on this insight, we further introduce an energy-domain minimum distance metric to quantify the distinguishability of different transmitted symbols after square-law detection. This metric provides a direct link between reference-wave design, RIS measurement diversity, and symbol detection performance.

The main contributions of this paper are summarized as follows:

- We establish a reference-assisted holographic RIS receiver model, in which complex-valued communication symbols are recovered from intensity-only measurements through the interference between the unknown signal and a known reference wave.
- We derive discrete identifiability conditions for reference-assisted phase retrieval over finite constellations and reveal the joint impact of the reference wave, the pulse-shaping matrix, and the RIS array manifold on symbol uniqueness.
- We introduce an energy-domain minimum distance metric to characterize the stability of symbol recovery and to guide reference-wave and RIS measurement design.
- We develop and compare maximum-likelihood and low-complexity phase retrieval based detectors, demonstrating that proper reference-wave diversity significantly improves symbol distinguishability and detection performance.

II. System Model

We consider a holographic RIS receiver system in which L single-antenna users transmit complex baseband symbols to an RIS composed of K reflecting elements. The RIS senses only the intensity of the interference pattern formed by the received signal and a known reference wave at each of its elements. The known reference signal is generated by a local oscillator [10]. The goal is to estimate the complex channel parameters and degree of arrival (DOA) angles, and subsequently recover the transmitted symbols utilizing these channel information.

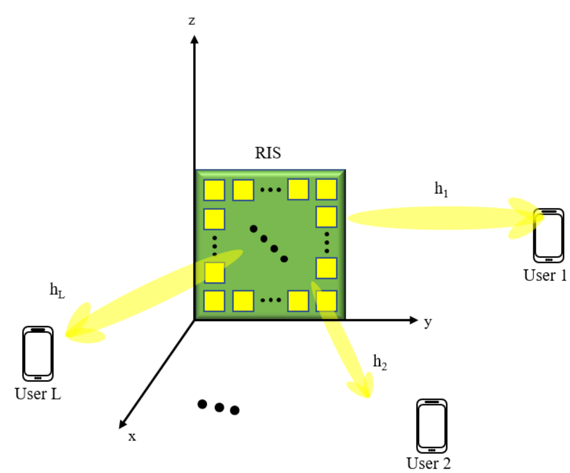

Fig. 1: System model of holographic RIS receiver with amplitude-only measurements.

A. Transmitter Model

Consider an uplink transmission system with L single-antenna users. The information-bearing symbols transmitted by the l-th user within one signaling block are collected as

$\mathbf{s}_l = [s_{l,0}, s_{l,1}, \ldots, s_{l,K-1}]^T \in \mathcal{S}^K$, where K denotes the number of symbols in the considered block, $s_{l,k}$ represents the k-th transmitted symbol of user l, and $\mathcal{S}$ denotes the modulation constellation. And the constellation symbols are assumed to be normalized as $\mathbb{E}\{|s_{l,k}|^2\} = 1$.

After pulse shaping, the complex baseband transmit waveform of the l-th user is given by

$$x_l(t) = \sqrt{P_l}\left(\sum_{k=0}^{K-1} s_{l,k} p(t - kT_s) + \xi_l q(t)\right), \tag{1}$$

where P_l is the average transmit power of user l, $p(t)$ is the pulse-shaping filter, T_s is the symbol duration, $\xi_l \in \mathbb{C}$ is a deterministic pilot or bias coefficient, and $q(t)$ is a known pilot waveform. The deterministic term $\xi_l q(t)$ is optional and is introduced to allow a nonzero-mean transmitted waveform when required by the subsequent reference-assisted recovery process. If no transmitted pilot or bias is used, we set $\xi_l = 0$.

The corresponding real-valued passband signal is expressed as

$$x_l^{\mathrm{RF}}(t) = \sqrt{2}\Re\left\{x_l(t)e^{j(2\pi f_c t + \varphi_l)}\right\}, \quad l = 1, 2, \ldots, L, \tag{2}$$

where f_c is the carrier frequency and φ_l denotes the initial carrier phase offset of the l-th user. Let

$$x_l(t) = x_{l,I}(t) + j x_{l,Q}(t), \tag{3}$$

where $x_{l,I}(t)$ and $x_{l,Q}(t)$ are the in-phase and quadrature components of the complex baseband waveform, respectively. Then (2) can be equivalently written as

$$\begin{aligned} x_l^{\mathrm{RF}}(t) = \sqrt{2}\big[&x_{l,I}(t)\cos(2\pi f_c t + \varphi_l) \\ &- x_{l,Q}(t)\sin(2\pi f_c t + \varphi_l)\big]. \end{aligned} \tag{4}$$

For compact notation, the multi-user symbol vector at symbol time k is defined as $\mathbf{s}_k = [s_{1,k}, s_{2,k}, \ldots, s_{L,k}]^T \in \mathcal{S}^L$, and the multi-user complex baseband waveform vector as $\mathbf{x}(t) = [x_1(t), x_2(t), \ldots, x_L(t)]^T \in \mathbb{C}^L$. Using (1), the multi-user baseband waveform can be written as

$$\mathbf{x}(t) = \mathbf{D}_P\left(\sum_{k=0}^{K-1} \mathbf{s}_k p(t - kT_s) + \boldsymbol{\xi} q(t)\right), \tag{5}$$

where $\mathbf{D}_P = \mathrm{diag}(\sqrt{P_1}, \sqrt{P_2}, \ldots, \sqrt{P_L})$ is the transmit-power scaling matrix, and $\boldsymbol{\xi} = [\xi_1, \xi_2, \ldots, \xi_L]^T \in \mathbb{C}^L$ collects the deterministic pilot or bias coefficients of all users.

To facilitate the subsequent symbol-level analysis, we further introduce an equivalent matrix representation for the pulse-shaping operation. Consider N_t baseband observation instants $\{t_n\}_{n=0}^{N_t-1}$ within the signaling block. For the l-th user, define the sampled baseband transmit vector as

$$\tilde{\mathbf{x}}_l = [x_l(t_0), x_l(t_1), \ldots, x_l(t_{N_t-1})]^T \in \mathbb{C}^{N_t}. \tag{6}$$

From (1), we obtain

$$\tilde{\mathbf{x}}_l = \sqrt{P_l}\left(\mathbf{P}\mathbf{s}_l + \xi_l \mathbf{q}\right), \tag{7}$$

where $\mathbf{P} \in \mathbb{C}^{N_t \times K}$ denotes the pulse-shaping matrix with entries $[\mathbf{P}]_{n,k} = p(t_n - kT_s)$ for $n = 0, 1, \ldots, N_t - 1$ and $k = 0, 1, \ldots, K-1$, and $\mathbf{q} = [q(t_0), q(t_1), \ldots, q(t_{N_t-1})]^T \in \mathbb{C}^{N_t}$ denotes the sampled pilot waveform.

Stacking the sampled baseband transmit vectors of all users gives

$$\tilde{\mathbf{x}} = [\tilde{\mathbf{x}}_1^T, \tilde{\mathbf{x}}_2^T, \ldots, \tilde{\mathbf{x}}_L^T]^T \in \mathbb{C}^{LN_t}. \tag{8}$$

Similarly, define the stacked transmitted symbol vector as

$$\mathbf{s} = [\mathbf{s}_1^T, \mathbf{s}_2^T, \ldots, \mathbf{s}_L^T]^T \in \mathcal{S}^{LK}. \tag{9}$$

Then the sampled multi-user transmit waveform can be represented as

$$\begin{aligned} \tilde{\mathbf{x}} &= (\mathbf{D}_P \otimes \mathbf{P})\mathbf{s} + (\mathbf{D}_P\boldsymbol{\xi}) \otimes \mathbf{q}, \\ &= \mathbf{P}_{\mathrm{tx}}\mathbf{s} + \mathbf{x}_0. \end{aligned} \tag{10}$$

where $\mathbf{P}_{\mathrm{tx}} = \mathbf{D}_P \otimes \mathbf{P} \in \mathbb{C}^{LN_t \times LK}$, $\mathbf{x}_0 = (\mathbf{D}_P\boldsymbol{\xi}) \otimes \mathbf{q} \in \mathbb{C}^{LN_t}$, and $\otimes$ denotes the Kronecker product. Thus, the transmitter-side equivalent discrete model is compactly written as

$$\tilde{\mathbf{x}} = \mathbf{P}_{\mathrm{tx}}\mathbf{s} + \mathbf{x}_0. \tag{11}$$

The matrix $\mathbf{P}_{\mathrm{tx}}$ characterizes the linear mapping from the transmitted constellation symbols to the sampled pulse-shaped waveforms. This representation allows the subsequent receiver model to incorporate pulse shaping, spatial propagation, and holographic RIS observations through an equivalent measurement matrix while keeping the analysis in the discrete symbol domain.

B. *Far-Field Channel Model*

As shown in Fig. 1, we consider a holographic RIS receiver composed of a uniform planar array placed on the yz-plane. The array consists of N_y elements along the y-axis and N_z elements along the z-axis, resulting in a total of

$$N_{\mathrm{r}} = N_y N_z \tag{12}$$

receiving elements. The position vector of the (n, m)-th element in the RIS array is defined as

$$\mathbf{r}_{n,m} = [0,\, y_n,\, z_m]^{\mathrm{T}}, \tag{13}$$

where $n = 1, \ldots, N_y$ and $m = 1, \ldots, N_z$, and $y_n = \left(n - \frac{N_y+1}{2}\right)d_y$, $z_m = \left(m - \frac{N_z+1}{2}\right)d_z$. Here, d_y and d_z denote the inter-element spacings along the y- and z-axes, respectively. The centered coordinate system is adopted for notational convenience; using a corner element as the origin leads only to a user-dependent common phase shift, which can be absorbed into the complex channel gain.

Under the far-field assumption, the impinging signal from each user can be modeled as a plane wave across the RIS aperture. Let θ_l and ϕ_l denote the angle of arrival (AoA) of the l-th user, where θ_l is the polar angle with respect to the array broadside, i.e., the positive x-axis, and ϕ_l is the azimuth angle in the yz-plane measured from the positive y-axis. The corresponding wave vector is given by

$$\mathbf{k}_l = [k_{l,x}, k_{l,y}, k_{l,z}]^T, \tag{14}$$

with $k_{l,x} = k_0\cos\theta_l$, $k_{l,y} = k_0\sin\theta_l\cos\phi_l$, $k_{l,z} = k_0\sin\theta_l\sin\phi_l$, $k_0 = \frac{2\pi}{\lambda}$ is the free-space wavenumber and λ is the carrier wavelength.

Following the plane-wave model, the phase of the incident field from user l at the (n, m)-th RIS element is determined by the projection of the wave vector onto the element position, i.e.,

$$\psi_{n,m}^{(l)} = \mathbf{k}_l^T \mathbf{r}_{n,m} = k_{l,y} y_n + k_{l,z} z_m. \tag{15}$$

Since the RIS aperture is assumed to be sufficiently small compared with the propagation distance, the amplitude variation of the incident wave across different RIS elements is neglected for each user. Therefore, the spatial variation across the RIS elements is mainly characterized by the phase profile in (15), while the large-scale path loss, shadowing, and user-dependent phase offset are incorporated into a complex channel gain.

Accordingly, the channel vector from the l-th user to all RIS receiving elements is modeled as

$$\mathbf{h}_l = \alpha_l \mathbf{v}_l \in \mathbb{C}^{N_\mathrm{r} \times 1}, \tag{16}$$

where $\alpha_l \in \mathbb{C}$ denotes the complex channel gain of user l, and $\mathbf{v}_l$ is the normalized array steering vector. The element of $\mathbf{v}_l$ corresponding to the (n, m)-th RIS element is given by

$$[\mathbf{v}_l]_{\iota(n,m)} = e^{j(k_{l,y} y_n + k_{l,z} z_m)}, \tag{17}$$

where $\iota(n, m) = (n-1)N_z + m$ maps the two-dimensional element index (n, m) to the one-dimensional vector index.

Equivalently, the array steering vector can be written as the Kronecker product of two one-dimensional steering vectors:

$$\mathbf{v}_l = \mathbf{a}_y(\theta_l, \phi_l) \otimes \mathbf{a}_z(\theta_l, \phi_l), \tag{18}$$

where $\otimes$ denotes the Kronecker product,

$$\mathbf{a}_y(\theta_l, \phi_l) = \left[e^{jk_{l,y}y_1}, e^{jk_{l,y}y_2}, \ldots, e^{jk_{l,y}y_{N_y}}\right]^T \in \mathbb{C}^{N_y \times 1}, \tag{19}$$

and

$$\mathbf{a}_z(\theta_l, \phi_l) = \left[e^{jk_{l,z}z_1}, e^{jk_{l,z}z_2}, \ldots, e^{jk_{l,z}z_{N_z}}\right]^T \in \mathbb{C}^{N_z \times 1}. \tag{20}$$

Collecting the channel vectors of all L users, the multi-user channel matrix between the users and the RIS receiver is expressed as

$$\begin{aligned} \mathbf{H} &= [\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_L] \in \mathbb{C}^{N_\mathrm{r} \times L} \\ &= [\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_L] \operatorname{diag}(\alpha_1, \alpha_2, \ldots, \alpha_L), \end{aligned} \tag{21}$$

where the second line follows from the decomposition of each user channel vector in (16). For compactness, define $\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_L] \in \mathbb{C}^{N_\mathrm{r} \times L}$ as the RIS array manifold matrix, and $\boldsymbol{\Gamma} = \operatorname{diag}(\alpha_1, \alpha_2, \ldots, \alpha_L) \in \mathbb{C}^{L \times L}$ as the diagonal matrix of complex channel gains. Then the multi-user channel matrix is compactly written as

$$\mathbf{H} = \mathbf{V}\boldsymbol{\Gamma}. \tag{22}$$

This decomposition separates the directional spatial response of the RIS, represented by $\mathbf{V}$, from the user-dependent complex propagation gains, represented by $\boldsymbol{\Gamma}$. It will be used in the subsequent receiver model to characterize how the pulse-shaped user signals are spatially mapped onto the holographic RIS elements.

C. Reference-Assisted Holographic Reception Model

At the receiver side, the holographic RIS collects the superimposed signals from all users through its N_r receiving elements. Using the multi-user channel matrix $\mathbf{H} \in \mathbb{C}^{N_\mathrm{r} \times L}$ defined in the previous subsection, the complex baseband signal impinging on the RIS at time t is written as

$$\mathbf{r}(t) = \mathbf{H}\mathbf{x}(t). \tag{23}$$

To enable phase information recovery from intensity-only observations, a known reference wave is introduced at the RIS receiver. Let $\mathbf{b} = [b_1, b_2, \ldots, b_{N_\mathrm{r}}]^T \in \mathbb{C}^{N_\mathrm{r}}$ denote the equivalent complex baseband reference vector, where $b_i = \rho_i e^{j\varphi_i}, i = 1, 2, \ldots, N_\mathrm{r}$. Here, ρ_i and φ_i are the known amplitude and phase of the reference wave injected at the i-th RIS element, respectively. The corresponding passband reference signal is

$$b_i^{\mathrm{RF}}(t) = \sqrt{2}\Re\left\{b_i e^{j2\pi f_c t}\right\} = \sqrt{2}\rho_i \cos(2\pi f_c t + \varphi_i). \tag{24}$$

After superimposing the received signal and the reference wave, the equivalent complex baseband field at the RIS is given by

$$\mathbf{u}(t) = \mathbf{r}(t) + \mathbf{b} = \mathbf{H}\mathbf{x}(t) + \mathbf{b}. \tag{25}$$

Equivalently, the real-valued RF signal at the i-th RIS element can be expressed as

$$u_i^{\mathrm{RF}}(t) = \sqrt{2}\Re\left\{u_i(t) e^{j2\pi f_c t}\right\} + \nu_i^{\mathrm{RF}}(t), \tag{26}$$

where $u_i(t)$ is the i-th element of $\mathbf{u}(t)$ and $\nu_i^{\mathrm{RF}}(t)$ denotes the RF noise.

The holographic RIS receiver is assumed to employ square-law detection at each element. After low-pass filtering or time averaging over a detection interval T_0, the intensity measurement at the i-th element is modeled as

$$z_i(t) = \frac{1}{T_0} \int_t^{t+T_0} \left|u_i^{\mathrm{RF}}(\tau)\right|^2 d\tau. \tag{27}$$

Under the narrowband and slowly varying envelope assumptions within the detection interval, the high-frequency components are removed by the low-pass operation, yielding the equivalent baseband intensity observation

$$z_i(t) = \left|[\mathbf{H}\mathbf{x}(t)]_i + b_i\right|^2 + w_i(t), \quad i = 1, 2, \ldots, N_\mathrm{r}, \tag{28}$$

where $w_i(t)$ denotes the effective post-detection noise and modeling error.

Stacking the observations of all RIS elements gives

$$\mathbf{z}(t) = \left|\mathbf{H}\mathbf{x}(t) + \mathbf{b}\right|^2 + \mathbf{w}(t), \tag{29}$$

where $\mathbf{z}(t) = [z_1(t), z_2(t), \ldots, z_{N_\mathrm{r}}(t)]^T \in \mathbb{R}^{N_\mathrm{r}}$, $\mathbf{w}(t) \in \mathbb{R}^{N_\mathrm{r}}$ is the effective intensity-domain noise vector, and the squared modulus in (29) is applied element-wise.

For subsequent symbol-level analysis, we evaluate (29) at the discrete observation instants $\{t_n\}_{n=0}^{N_t-1}$. Let

$$\mathbf{x}[n] \triangleq \mathbf{x}(t_n), \quad \mathbf{z}[n] \triangleq \mathbf{z}(t_n), \quad \mathbf{w}[n] \triangleq \mathbf{w}(t_n). \tag{30}$$

Then the discrete-time reference-assisted intensity measurement is

$$\mathbf{z}[n] = \left|\mathbf{H}\mathbf{x}[n] + \mathbf{b}\right|^2 + \mathbf{w}[n], \quad n = 0, 1, \ldots, N_t - 1. \tag{31}$$

Finally, using the transmitter-side equivalent matrix model in (11), the complete block-level observation can be compactly represented as

$$\mathbf{z} = |\mathbf{G}\mathbf{s} + \mathbf{b}_{\mathrm{eq}}|^2 + \mathbf{w}, \tag{32}$$

where $\mathbf{s} \in \mathcal{S}^{LK}$ is the stacked transmitted symbol vector, $\mathbf{G}$ is the equivalent measurement matrix incorporating pulse shaping, propagation, and RIS spatial responses, $\mathbf{b}_{\mathrm{eq}}$ is the stacked equivalent reference vector, and $\mathbf{z}$ and $\mathbf{w}$ collect all intensity observations and post-detection noise samples, respectively. The compact model in (32) will serve as the basis for the subsequent identifiability analysis and detector design.

D. Two-Phase Transmission Protocol and Unified Observation Model

Based on the reference-assisted holographic reception model, we consider a two-phase transmission protocol consisting of a training phase and a data transmission phase. In the training phase, the users transmit known pilot waveforms, and the RIS receiver estimates the channel from intensity-only measurements. In the data transmission phase, the users transmit information-bearing symbols, and the receiver detects the transmitted symbols based on the estimated channel. Both phases follow the same reference-assisted intensity observation principle, but the unknown variables are different.

1) Training Phase: During the training phase, all users transmit known pilot signals. Let $\mathbf{x}_{\mathrm{p}}[t] \in \mathbb{C}^{L}$ denote the known multi-user pilot vector at the t-th pilot observation, where $t = 0, 1, \ldots, T_{\mathrm{p}}-1$ and T_{p} is the number of pilot observations. The corresponding reference vector is denoted by $\mathbf{b}_{\mathrm{p}}[t] \in \mathbb{C}^{N_{\mathrm{r}}}$. According to (31), the received intensity measurement during the training phase for time instants $t = 0, 1, \ldots, T_{\mathrm{p}} - 1$ is given by

$$\mathbf{z}_{\mathrm{p}}[t] = |\mathbf{H}\mathbf{x}_{\mathrm{p}}[t] + \mathbf{b}_{\mathrm{p}}[t]|^2 + \mathbf{w}_{\mathrm{p}}[t]. \tag{33}$$

where $\mathbf{z}_{\mathrm{p}}[t] \in \mathbb{R}^{N_{\mathrm{r}}}$ is the intensity observation vector, and $\mathbf{w}_{\mathrm{p}}[t] \in \mathbb{R}^{N_{\mathrm{r}}}$ denotes the effective post-detection noise.

For compact representation, define the vectorized channel as

$$\mathbf{h} = \mathrm{vec}(\mathbf{H}) \in \mathbb{C}^{N_{\mathrm{r}}L}. \tag{34}$$

Using the identity

$$\mathbf{H}\mathbf{x}_{\mathrm{p}}[t] = \left(\mathbf{x}_{\mathrm{p}}^{T}[t] \otimes \mathbf{I}_{N_{\mathrm{r}}}\right)\mathbf{h}, \tag{35}$$

(33) can be rewritten as

$$\mathbf{z}_{\mathrm{p}}[t] = |\mathbf{G}_{\mathrm{p}}[t]\mathbf{h} + \mathbf{b}_{\mathrm{p}}[t]|^2 + \mathbf{w}_{\mathrm{p}}[t], \tag{36}$$

where $\mathbf{G}_{\mathrm{p}}[t] = \mathbf{x}_{\mathrm{p}}^{T}[t] \otimes \mathbf{I}_{N_{\mathrm{r}}} \in \mathbb{C}^{N_{\mathrm{r}} \times N_{\mathrm{r}}L}$. Stacking all pilot observations yields the block-level training model

$$\mathbf{z}_{\mathrm{p}} = |\mathbf{G}_{\mathrm{p}}\mathbf{h} + \mathbf{b}_{\mathrm{p}}|^2 + \mathbf{w}_{\mathrm{p}}, \tag{37}$$

where $\mathbf{z}_{\mathrm{p}}$, $\mathbf{b}_{\mathrm{p}}$, and $\mathbf{w}_{\mathrm{p}}$ collect the corresponding quantities over all pilot observations, and $\mathbf{G}_{\mathrm{p}} = \left[\mathbf{G}_{\mathrm{p}}[0], \mathbf{G}_{\mathrm{p}}[1], \ldots, \mathbf{G}_{\mathrm{p}}[T_{\mathrm{p}} - 1]\right]^{\mathrm{T}}$. If the far-field channel structure in (22) is exploited, the unknown channel vector $\mathbf{h}$ can be further parameterized by a lower-dimensional channel parameter vector, such as the complex path-gain vector when the angles of arrival are known or pre-estimated.

2) Data Transmission Phase: After the training phase, the receiver obtains an estimate of the channel, denoted by $\hat{\mathbf{H}}$. In the data transmission phase, the users transmit unknown information-bearing symbols. Let $\mathbf{s}_{\mathrm{d}} \in \mathcal{S}^{LK_{\mathrm{d}}}$ denote the stacked data-symbol vector within one data block, where K_{d} is the number of data symbols per user. Following the transmitter-side equivalent model, the sampled data waveform can be expressed as

$$\tilde{\mathbf{x}}_{\mathrm{d}} = \mathbf{P}_{\mathrm{tx},d}\mathbf{s}_{\mathrm{d}} + \mathbf{x}_{0,d}, \tag{38}$$

where $\mathbf{P}_{\mathrm{tx},d}$ is the data-phase pulse-shaping matrix and $\mathbf{x}_{0,d}$ is the deterministic transmit-side pilot or bias component, if present.

Using the estimated channel $\hat{\mathbf{H}}$, the block-level data-phase intensity observation can be written as

$$\mathbf{z}_{\mathrm{d}} = \left|\mathbf{G}_{\mathrm{d}}(\hat{\mathbf{H}})\mathbf{s}_{\mathrm{d}} + \mathbf{b}_{\mathrm{d,eq}}\right|^2 + \mathbf{w}_{\mathrm{d}}, \tag{39}$$

where $\mathbf{G}_{\mathrm{d}}(\hat{\mathbf{H}})$ is the equivalent data-phase measurement matrix determined by the estimated channel, the pulse-shaping matrix, and the RIS observation structure. The vector $\mathbf{b}_{\mathrm{d,eq}}$ denotes the equivalent known offset in the data phase, which includes the receiver-side reference wave and the contribution of any deterministic transmit-side bias. The vector $\mathbf{w}_{\mathrm{d}}$ represents the effective post-detection noise.

Equations (37) and (39) show that both channel estimation and data detection can be written in the common reference-assisted intensity measurement form

$$\mathbf{z} = |\mathbf{G}\boldsymbol{\theta} + \mathbf{b}_{\mathrm{eq}}|^2 + \mathbf{w}. \tag{40}$$

In the training phase, the unknown variable is the channel-related parameter, e.g., $\boldsymbol{\theta} = \mathbf{h}$ or a lower-dimensional parameterization of $\mathbf{H}$. In the data transmission phase, the unknown variable is the finite-alphabet symbol vector, i.e., $\boldsymbol{\theta} = \mathbf{s}_{\mathrm{d}}$. Therefore, the two phases share the same phaseless observation structure, while their identifiability conditions and recovery algorithms differ due to the different structures of the unknown variables. The unified model in (40) will serve as the basis for the subsequent identifiability analysis and detector design.

III. Uniqueness and Stability of Reference-Assisted Energy Detection

This section studies when the reference-assisted square-law receiver has a unique inverse and when that inverse is robust to perturbations. Following the distinction between injectivity and stability in phase retrieval [?], we do not use matrix rank alone as a surrogate for recoverability. Instead, we derive an exact pairwise condition and lower and upper Lipschitz bounds for the nonlinear energy mapping. The analysis applies to both continuous channel estimation and finite-alphabet symbol detection.

A. Unified Affine Phase-Retrieval Model

The training- and data-phase observations derived in Section II share the form

$$\mathbf{z} = \mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}) + \mathbf{w}, \qquad \mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}) \triangleq |\mathbf{G}\boldsymbol{\theta} + \mathbf{b}|^2, \tag{41}$$

where $\mathbf{G} \in \mathbb{C}^{Q\times D}$, $\boldsymbol{\theta} \in \mathcal{X} \subseteq \mathbb{C}^D$, $\mathbf{b} \in \mathbb{C}^Q$, and the modulus and square are element-wise. This is an affine phase-retrieval model [9]. In the training phase, $\boldsymbol{\theta} = \mathbf{h} = \mathrm{vec}(\mathbf{H})$, or a lower dimensional channel parameter; in the data phase, $\boldsymbol{\theta} = \mathbf{s}_\mathrm{d} \in \mathcal{S}^{LK_\mathrm{d}}$.

The reference term changes the ambiguity class fundamentally. If $\mathbf{b} = 0$, then

$$\mathcal{T}_{\mathbf{G},\mathbf{0}}(e^{j\varphi}\boldsymbol{\theta}) = \mathcal{T}_{\mathbf{G},\mathbf{0}}(\boldsymbol{\theta}), \tag{42}$$

and uniqueness can only be defined modulo a global phase. If $\mathbf{b} \neq 0$, the expansion

$$\mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}) = |\mathbf{G}\boldsymbol{\theta}|^2 + |\mathbf{b}|^2 + 2\,\mathrm{Re}\{\mathbf{b}^* \odot \mathbf{G}\boldsymbol{\theta}\} \tag{43}$$

contains phase-sensitive interference, so absolute uniqueness on $\mathcal{X}$ becomes possible. A nonzero reference is not sufficient by itself, however: its phases must probe every feasible field difference in nondegenerate real directions.

To quantify this statement, define the realification

$$\boldsymbol{\theta}_\mathbb{R} \triangleq \left[\mathrm{Re}\{\boldsymbol{\theta}\}^T \quad \mathrm{Im}\{\boldsymbol{\theta}\}^T\right]^T \in \mathbb{R}^{2D} \tag{44}$$

and, for $\boldsymbol{\theta}_1 \neq \boldsymbol{\theta}_2$, the energy-domain distance

$$d_E(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2) \triangleq \left\|\mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}_1) - \mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}_2)\right\|_2. \tag{45}$$

The optimal lower and upper stability bounds on $\mathcal{X}$ are

$$\alpha_\mathcal{X} \triangleq \inf_{\substack{\boldsymbol{\theta}_1,\boldsymbol{\theta}_2\in\mathcal{X}\\ \boldsymbol{\theta}_1\neq\boldsymbol{\theta}_2}} \frac{d_E(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2)}{\|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|_2}, \tag{46}$$

$$\beta_\mathcal{X} \triangleq \sup_{\substack{\boldsymbol{\theta}_1,\boldsymbol{\theta}_2\in\mathcal{X}\\ \boldsymbol{\theta}_1\neq\boldsymbol{\theta}_2}} \frac{d_E(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2)}{\|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|_2}. \tag{47}$$

Thus, the inverse problem is globally stable on $\mathcal{X}$ if $0 < \alpha_\mathcal{X} \leq \beta_\mathcal{X} < \infty$. The lower bound is the important one: it excludes pairs that are far apart in parameter space but arbitrarily close in energy space.

B. Exact Secant Identity and Global Uniqueness

The quadratic structure of (41) gives an exact identity, which is stronger than a first-order approximation. For a point $\boldsymbol{\vartheta} \in \mathbb{C}^D$, define the real Jacobian

$$\begin{aligned}\mathbf{J}(\boldsymbol{\vartheta}) \triangleq 2\big[&\mathrm{Re}\{\mathrm{diag}[(\mathbf{G}\boldsymbol{\vartheta} + \mathbf{b})^*]\mathbf{G}\},\\ &-\mathrm{Im}\{\mathrm{diag}[(\mathbf{G}\boldsymbol{\vartheta} + \mathbf{b})^*]\mathbf{G}\}\big] \in \mathbb{R}^{Q\times 2D}.\end{aligned} \tag{48}$$

Lemma 1 (Exact midpoint factorization)**.** *For any $\boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \mathbb{C}^D$, let $\boldsymbol{\mu} = (\boldsymbol{\theta}_1 + \boldsymbol{\theta}_2)/2$ and $\boldsymbol{\delta} = \boldsymbol{\theta}_1 - \boldsymbol{\theta}_2$. Then*

$$\mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}_1) - \mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}_2) = \mathbf{J}(\boldsymbol{\mu})\boldsymbol{\delta}_\mathbb{R}. \tag{49}$$

Proof. Applying $|a|^2 - |c|^2 = \mathrm{Re}\{(a + c)^*(a - c)\}$ element-wise gives

$$\mathrm{Re}\big\{[\mathbf{G}(\boldsymbol{\theta}_1 + \boldsymbol{\theta}_2) + 2\mathbf{b}]^* \odot \mathbf{G}(\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2)\big\}, \tag{50}$$

which equals the right-hand side of (49) after realification. □

Theorem 1 (Pairwise uniqueness criterion)**.** *The mapping $\mathcal{T}_{\mathbf{G},\mathbf{b}}$ is injective on $\mathcal{X}$ if and only if, for every distinct $\boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \mathcal{X}$,*

$$\mathbf{J}\left(\frac{\boldsymbol{\theta}_1 + \boldsymbol{\theta}_2}{2}\right)(\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2)_\mathbb{R} \neq \mathbf{0}. \tag{51}$$

A necessary condition, independent of the reference, is

$$\begin{aligned}&\mathrm{Null}(\mathbf{G}) \cap \mathcal{D}_\mathcal{X} = \varnothing,\\ &\mathcal{D}_\mathcal{X} \triangleq \{\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2 : \boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \mathcal{X},\ \boldsymbol{\theta}_1 \neq \boldsymbol{\theta}_2\}.\end{aligned} \tag{52}$$

Proof. The first claim follows directly from Lemma 1. For the second, if $\mathbf{G}(\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2) = 0$, then the two complex fields, and hence their intensities for every $\mathbf{b}$, are identical. □

Theorem 1 makes clear why $\mathrm{rank}(\mathbf{G}) = D$ is necessary for a full-dimensional continuous parameter set but is not sufficient for phase retrieval: even when $\mathbf{G}\boldsymbol{\delta} \neq 0$, its real projection determined jointly by the midpoint field and the reference may vanish. For a finite constellation, full column rank is not necessary; only the finite difference set in (52) must avoid the null space.

Let

$$\mathcal{M}_\mathcal{X} \triangleq \{(\boldsymbol{\theta}_1 + \boldsymbol{\theta}_2)/2 : \boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \mathcal{X}\} \tag{53}$$

be the midpoint set. The exact identity immediately yields computable two-sided bounds.

Proposition 1 (Jacobian stability bounds)**.** *Define*

$$\underline{\sigma}_J \triangleq \inf_{\boldsymbol{\mu}\in\mathcal{M}_\mathcal{X}} \sigma_{\min}(\mathbf{J}(\boldsymbol{\mu})), \qquad \overline{\sigma}_J \triangleq \sup_{\boldsymbol{\mu}\in\mathcal{M}_\mathcal{X}} \sigma_{\max}(\mathbf{J}(\boldsymbol{\mu})). \tag{54}$$

Then

$$\underline{\sigma}_J \,\|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|_2 \leq d_E(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2) \leq \overline{\sigma}_J \,\|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|_2 \tag{55}$$

for all pairs in $\mathcal{X}$. Consequently, $\alpha_\mathcal{X} \geq \underline{\sigma}_J$ and $\beta_\mathcal{X} \leq \overline{\sigma}_J$. If $\|\boldsymbol{\theta}\|_2 \leq R_\mathcal{X}$ on $\mathcal{X}$, then

$$\beta_\mathcal{X} \leq 2\|\mathbf{G}\|_2 \left(\|\mathbf{G}\|_2 R_\mathcal{X} + \|\mathbf{b}\|_\infty\right). \tag{56}$$

Proof. The first statement follows by applying the singular-value inequalities to (49). For the explicit upper bound, use $\|\mathrm{diag}(\mathbf{u}^*)\mathbf{G}\|_2 \leq \|\mathbf{u}\|_\infty \|\mathbf{G}\|_2$ and $\|\mathbf{G}\boldsymbol{\mu}+\mathbf{b}\|_\infty \leq \|\mathbf{G}\|_2 R_\mathcal{X} + \|\mathbf{b}\|_\infty$. □

The bound $\underline{\sigma}_J > 0$ is a convenient sufficient condition for global stability, but it can be conservative because it tests every direction at every midpoint, including directions that may not connect two feasible points. The exact constant in (46), or equivalently the restricted secants in (49), is the sharp criterion. Notice also that injectivity alone need not give a useful uniform lower bound on a continuous set; this is precisely the distinction between uniqueness and stability emphasized in [**?**].

C. Continuous Channel Identifiability

For a continuous channel parameter, local identifiability at $\boldsymbol{\theta}$ requires

$$\mathrm{rank}(\mathbf{J}(\boldsymbol{\theta})) = 2D, \qquad Q \geq 2D. \tag{57}$$

The smallest singular value determines the local inverse sensitivity:

$$\liminf_{\|\boldsymbol{\delta}\|_2\to 0} \frac{d_E(\boldsymbol{\theta} + \boldsymbol{\delta}, \boldsymbol{\theta})}{\|\boldsymbol{\delta}\|_2} = \sigma_{\min}(\mathbf{J}(\boldsymbol{\theta})). \tag{58}$$

Local full rank does not rule out a distant ambiguous channel. Global uniqueness still requires (51) for all pairs, while a positive $\alpha_\mathcal{X}$ rules out both exact and near-ambiguities.

These constants also have a direct error interpretation. Let $\boldsymbol{\theta}_\star \in \mathcal{X}$ generate (41), and let $\widehat{\boldsymbol{\theta}}$ be a global minimizer of the intensity least-squares criterion. If $\alpha_{\mathcal{X}} > 0$, then

$$\|\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}_\star\|_2 \le \frac{2\|\mathbf{w}\|_2}{\alpha_{\mathcal{X}}}. \tag{59}$$

Indeed, optimality gives a residual no larger than $\|\mathbf{w}\|_2$; the triangle inequality followed by the lower Lipschitz bound gives (59).

For full-channel training,

$$\mathbf{G}_{\mathrm{p}} = \mathbf{X}_{\mathrm{p}} \otimes \mathbf{I}_{N_{\mathrm{r}}}, \qquad \mathbf{X}_{\mathrm{p}} \in \mathbb{C}^{T_{\mathrm{p}} \times L}, \tag{60}$$

and therefore

$$\begin{aligned} \operatorname{rank}(\mathbf{G}_{\mathrm{p}}) &= N_{\mathrm{r}} \operatorname{rank}(\mathbf{X}_{\mathrm{p}}), \\ \sigma_{\min}(\mathbf{G}_{\mathrm{p}}) &= \sigma_{\min}(\mathbf{X}_{\mathrm{p}}), \\ \kappa(\mathbf{G}_{\mathrm{p}}) &= \kappa(\mathbf{X}_{\mathrm{p}}). \end{aligned} \tag{61}$$

Thus, $\operatorname{rank}(\mathbf{X}_{\mathrm{p}}) = L$ and $T_{\mathrm{p}} \ge L$ are necessary to preserve the complex channel field. They are not sufficient for the intensity mapping. With one real intensity per RIS element and pilot time, $Q = N_{\mathrm{r}} T_{\mathrm{p}}$ and $D = N_{\mathrm{r}} L$, so the dimension condition in (57) strengthens the counting requirement to

$$T_{\mathrm{p}} \ge 2L. \tag{62}$$

Even when this count is met, the pilot and reference phases must make $\mathbf{J}_{\mathrm{p}}$ full column rank and well conditioned.

If the far-field model $\mathbf{H} = \mathbf{V}\boldsymbol{\Gamma}$ is used and the angles are known, the unknown can be reduced to $\boldsymbol{\alpha} = [\alpha_1, \ldots, \alpha_L]^T$, with

$$\mathbf{G}_\alpha = \begin{bmatrix} \mathbf{V} \operatorname{diag}(\mathbf{x}_{\mathrm{p}}[0]) \\ \vdots \\ \mathbf{V} \operatorname{diag}(\mathbf{x}_{\mathrm{p}}[T_{\mathrm{p}} - 1]) \end{bmatrix}. \tag{63}$$

The necessary count becomes $N_{\mathrm{r}} T_{\mathrm{p}} \ge 2L$. In addition, $\mathbf{G}_\alpha$ must preserve every gain direction and the reference must provide two independent real projections of those directions. Closely spaced angles reduce $\sigma_{\min}(\mathbf{V})$ and hence both the field-domain and energy-domain stability margins.

D. Finite-Alphabet Uniqueness and Noise Margin

For data detection, let $\mathcal{X}_{\mathrm{d}} = \mathcal{S}^{LK_{\mathrm{d}}}$. Since this set is finite, global uniqueness and a positive minimum separation are equivalent. Define

$$d_{\min,\mathrm{d}}^{(E)} \triangleq \min_{\substack{\mathbf{s}_1, \mathbf{s}_2 \in \mathcal{X}_{\mathrm{d}} \\ \mathbf{s}_1 \ne \mathbf{s}_2}} \left\| |\mathbf{G}_{\mathrm{d}}\mathbf{s}_1 + \mathbf{b}_{\mathrm{d}}|^2 - |\mathbf{G}_{\mathrm{d}}\mathbf{s}_2 + \mathbf{b}_{\mathrm{d}}|^2 \right\|_2. \tag{64}$$

Corollary 1 (Discrete uniqueness and stability)**.** *The symbols are uniquely identifiable if and only if $d_{\min,\mathrm{d}}^{(E)} > 0$. Equivalently, for every distinct pair,*

$$\mathbf{J}_{\mathrm{d}}\left(\frac{\mathbf{s}_1 + \mathbf{s}_2}{2}\right)(\mathbf{s}_1 - \mathbf{s}_2)_{\mathbb{R}} \ne \mathbf{0}. \tag{65}$$

Moreover, the nearest-intensity detector is guaranteed to return the transmitted symbol vector whenever

$$\|\mathbf{w}\|_2 < \frac{d_{\min,\mathrm{d}}^{(E)}}{2}. \tag{66}$$

For additive white Gaussian post-detection noise with covariance $\sigma_w^2\mathbf{I}$, the pairwise ML error probability is

$$P(\mathbf{s}_1 \to \mathbf{s}_2) = Q\left(\frac{d_E(\mathbf{s}_1, \mathbf{s}_2)}{2\sigma_w}\right), \tag{67}$$

so (64) is both a uniqueness certificate and the worst-case noise margin.

The data-phase field matrix has the factorization

$$\mathbf{G}_{\mathrm{d}} = (\mathbf{I}_{N_t} \otimes \widehat{\mathbf{H}})\boldsymbol{\Pi}\mathbf{P}_{\mathrm{tx},d}. \tag{68}$$

When the factors have full column rank,

$$\sigma_{\min}(\mathbf{G}_{\mathrm{d}}) \ge \sigma_{\min}(\widehat{\mathbf{H}})\sigma_{\min}(\mathbf{P}_{\mathrm{tx},\mathrm{d}}), \tag{69}$$

$$\sigma_{\min}(\widehat{\mathbf{H}}) \approx \sigma_{\min}(\mathbf{V}\boldsymbol{\Gamma}) \ge \sigma_{\min}(\mathbf{V}) \min_l |\alpha_l|. \tag{70}$$

These inequalities quantify only preservation in the complex field domain. Theorem 1 adds the missing condition: the reference-induced real projection must also be nonzero for every constellation secant.

E. A Constructive Four-Phase Reference Bound

The preceding conditions are exact but depend on the unknown midpoint. A phase-complete reference pattern gives a simple midpoint-independent sufficient condition. Suppose a base field sample $u_q = [\mathbf{G}\boldsymbol{\theta}]_q$ is repeated with four known references

$$b_{q,k} = \rho_q e^{j\phi_k}, \qquad \phi_k \in \{0, \pi/2, \pi, 3\pi/2\}, \quad \rho_q > 0. \tag{71}$$

Denote the resulting stacked mapping by $\mathcal{T}_{4\mathrm{ph}}(\boldsymbol{\theta})$.

Theorem 2 (Four-phase global lower bound)**.** *For any $\boldsymbol{\theta}_1, \boldsymbol{\theta}_2$,*

$$\begin{aligned} &\|\mathcal{T}_{4\mathrm{ph}}(\boldsymbol{\theta}_1) - \mathcal{T}_{4\mathrm{ph}}(\boldsymbol{\theta}_2)\|_2^2 \\ &\quad = 4\left\||\mathbf{G}\boldsymbol{\theta}_1|^2 - |\mathbf{G}\boldsymbol{\theta}_2|^2\right\|_2^2 + 8\sum_q \rho_q^2 |[\mathbf{G}(\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2)]_q|^2. \end{aligned} \tag{72}$$

Consequently, with $\rho_{\min} = \min_q \rho_q$,

$$\|\mathcal{T}_{4\mathrm{ph}}(\boldsymbol{\theta}_1) - \mathcal{T}_{4\mathrm{ph}}(\boldsymbol{\theta}_2)\|_2 \ge 2\sqrt{2}\rho_{\min}\|\mathbf{G}(\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2)\|_2. \tag{73}$$

If $\mathbf{G}$ has full column rank, then

$$\alpha_{\mathcal{X}} \ge 2\sqrt{2}\rho_{\min}\sigma_{\min}(\mathbf{G}) > 0 \tag{74}$$

for every feasible set $\mathcal{X}$. For a finite alphabet, full column rank can be replaced by $\operatorname{Null}(\mathbf{G}) \cap \mathcal{D}_{\mathcal{X}} = \varnothing$.

Proof. For one field coordinate, put $c_q = |u_{1,q}|^2 - |u_{2,q}|^2$ and $\Delta u_q = u_{1,q} - u_{2,q}$. Summing $[c_q + 2\rho_q \operatorname{Re}\{e^{-j\phi_k}\Delta u_q\}]^2$ over the four phases cancels all cross terms and gives $4c_q^2 + 8\rho_q^2|\Delta u_q|^2$. Summing over q proves (72); the remaining claims follow from singular value or restricted-difference bounds. □

The construction also has a transparent holographic interpretation. Opposite reference phases cancel the unknown self-intensity:

$$\operatorname{Re}\{u_q\} = \frac{z_{q,0} - z_{q,\pi}}{4\rho_q}, \quad \operatorname{Im}\{u_q\} = \frac{z_{q,\pi/2} - z_{q,3\pi/2}}{4\rho_q}. \tag{75}$$

Thus, four phase-shifted energy measurements recover two orthogonal real projections of each complex field sample. This proves explicitly that reference phase diversity can remove the global phase ambiguity and yields the stability lower bound in (74). Fewer or hardware-constrained phases may still be unique, but their quality should be evaluated through $\alpha_{\mathcal{X}}$, $\underline{\sigma}_J$, or $d^{(E)}_{\min,\mathrm{d}}$, rather than through the rank of $\mathbf{G}$ alone.

The resulting design criteria are therefore

$$\max_{\mathbf{b}_\mathrm{p}\in\mathcal{B}_\mathrm{p}} \inf_{\boldsymbol{\theta}\in\mathcal{X}_\mathrm{p}} \sigma_{\min}[\mathbf{J}_\mathrm{p}(\boldsymbol{\theta})] \quad \text{and} \quad \max_{\mathbf{b}_\mathrm{d}\in\mathcal{B}_\mathrm{d}} d^{(E)}_{\min,\mathrm{d}} \tag{76}$$

for continuous training and discrete data detection, respectively. The first maximizes a uniform local/global lower bound, whereas the second directly maximizes the finite-alphabet decision margin.

IV. Estimation and Detection Algorithms

Based on the reference-assisted observation model and reference-wave design principles developed in the previous sections, this section presents practical estimation and detection algorithms for the proposed holographic RIS receiver. The goal of this section is not to introduce a new family of generic phase retrieval algorithms, but to show how the proposed two-phase communication framework can be implemented using channel estimation and symbol detection procedures that are consistent with the identifiability analysis.

A. Overall Two-Phase Recovery Framework

The proposed receiver operates in two phases: a training phase for channel estimation and a data transmission phase for symbol detection. In the training phase, known pilot symbols and designed reference waves are used to estimate the channel-related parameter. In the data phase, the estimated channel is used to construct the equivalent measurement matrix for detecting the finite-alphabet data symbols from reference-assisted intensity measurements.

The overall recovery procedure is summarized in Algorithm 1. First, the receiver selects or generates a training reference vector $\mathbf{b}_\mathrm{p}$ according to the design criteria discussed in Section IV. The users then transmit pilot signals, and the RIS receiver collects the training intensity measurements $\mathbf{z}_\mathrm{p}$. Based on the training model

$$\mathbf{z}_\mathrm{p} = |\mathbf{G}_\mathrm{p}\boldsymbol{\theta}_\mathrm{c} + \mathbf{b}_\mathrm{p}|^2 + \mathbf{w}_\mathrm{p}, \tag{77}$$

the receiver estimates the channel parameter $\boldsymbol{\theta}_\mathrm{c}$ and reconstructs the channel estimate $\hat{\mathbf{H}}$.

After channel estimation, the data-phase reference vector $\mathbf{b}_\mathrm{d}$ is selected. The receiver then forms the data-phase equivalent measurement matrix $\mathbf{G}_\mathrm{d}(\hat{\mathbf{H}})$ according to the estimated channel and the transmit-side pulse-shaping model. Given the data-phase intensity measurements

$$\mathbf{z}_\mathrm{d} = \left|\mathbf{G}_\mathrm{d}(\hat{\mathbf{H}})\mathbf{s}_\mathrm{d} + \mathbf{b}_\mathrm{d}\right|^2 + \mathbf{w}_\mathrm{d}, \tag{78}$$

the receiver detects the transmitted symbol vector $\mathbf{s}_\mathrm{d}$ using either maximum-likelihood detection or a low-complexity phase-retrieval-based detector.

The two phases share the same reference-assisted phase retrieval structure, but they differ in the feasible set of the unknown variable. In the training phase, the unknown channel parameter is continuous, and the recovery performance is closely related to the rank and conditioning of the Jacobian of the intensity mapping. In the data phase, the unknown symbol vector belongs to a finite constellation set, and the detection performance is governed by the energy-domain minimum distance. Therefore, the subsequent subsections present tailored algorithms for channel estimation and symbol detection, respectively.

Algorithm 1 Two-Phase Reference-Assisted Holographic Reception

Require: Pilot sequences, constellation set $\mathcal{S}$, RIS geometry, pulse-shaping matrix, reference codebooks or reference design rules.
Ensure: Estimated channel $\hat{\mathbf{H}}$ and detected data symbols $\hat{\mathbf{s}}_\mathrm{d}$.

Training phase:
Select the training reference vector $\mathbf{b}_\mathrm{p}$ according to the channel-estimation design criterion.
Users transmit known pilot signals.
RIS receiver collects intensity measurements $\mathbf{z}_\mathrm{p}$.
Estimate the channel-related parameter $\hat{\boldsymbol{\theta}}_\mathrm{c}$ from

$$\mathbf{z}_\mathrm{p} = |\mathbf{G}_\mathrm{p}\boldsymbol{\theta}_\mathrm{c} + \mathbf{b}_\mathrm{p}|^2 + \mathbf{w}_\mathrm{p}.$$

Reconstruct the channel estimate $\hat{\mathbf{H}}$ from $\hat{\boldsymbol{\theta}}_\mathrm{c}$.
Data transmission phase:
Select the data-phase reference vector $\mathbf{b}_\mathrm{d}$ according to the symbol-detection design criterion.
Construct the equivalent data-phase measurement matrix $\mathbf{G}_\mathrm{d}(\hat{\mathbf{H}})$.
Users transmit data symbols.
RIS receiver collects intensity measurements $\mathbf{z}_\mathrm{d}$.
Detect the data-symbol vector from

$$\mathbf{z}_\mathrm{d} = \left|\mathbf{G}_\mathrm{d}(\hat{\mathbf{H}})\mathbf{s}_\mathrm{d} + \mathbf{b}_\mathrm{d}\right|^2 + \mathbf{w}_\mathrm{d}.$$

Output $\hat{\mathbf{H}}$ and $\hat{\mathbf{s}}_\mathrm{d}$.

B. Reference-Assisted Channel Estimation

We first consider the channel estimation problem in the training phase. Given the designed training reference vector $\mathbf{b}_\mathrm{p}$ and the known pilot matrix, the receiver estimates the channel-related parameter from reference-assisted intensity measurements. According to the training model, the observation can be written as

$$\mathbf{z}_\mathrm{p} = |\mathbf{G}_\mathrm{p}\boldsymbol{\theta}_\mathrm{c} + \mathbf{b}_\mathrm{p}|^2 + \mathbf{w}_\mathrm{p}, \tag{79}$$

where $\boldsymbol{\theta}_\mathrm{c} \in \mathbb{C}^{D_\mathrm{c}}$ denotes the channel-related unknown parameter. For full channel estimation, $\boldsymbol{\theta}_\mathrm{c} = \mathrm{vec}(\mathbf{H})$. When the far-field structure $\mathbf{H} = \mathbf{V}\boldsymbol{\Gamma}$ is exploited and the angles of arrival are known or pre-estimated, $\boldsymbol{\theta}_\mathrm{c}$ can be reduced to the complex path-gain vector $\boldsymbol{\alpha}$.

The channel estimation problem is formulated as a nonlinear least-squares problem:

$$\hat{\boldsymbol{\theta}}_\mathrm{c} = \arg\min_{\boldsymbol{\theta}_\mathrm{c}\in\mathbb{C}^{D_\mathrm{c}}} \left\|\mathbf{z}_\mathrm{p} - |\mathbf{G}_\mathrm{p}\boldsymbol{\theta}_\mathrm{c} + \mathbf{b}_\mathrm{p}|^2\right\|_2^2. \tag{80}$$

Since the objective function is nonconvex due to the square-law operation, we solve (80) using a damped Gauss–Newton method. This choice is consistent with the local identifiability analysis in Section III, where the Jacobian of the intensity mapping plays a central role.

Let

$$\mathbf{u}_{\mathrm{p}}^{(t)} = \mathbf{G}_{\mathrm{p}}\boldsymbol{\theta}_{\mathrm{c}}^{(t)} + \mathbf{b}_{\mathrm{p}} \tag{81}$$

be the predicted complex field at the t-th iteration, and define the predicted intensity as

$$\hat{\mathbf{z}}_{\mathrm{p}}^{(t)} = \left|\mathbf{u}_{\mathrm{p}}^{(t)}\right|^2. \tag{82}$$

The residual vector is

$$\mathbf{e}^{(t)} = \mathbf{z}_{\mathrm{p}} - \hat{\mathbf{z}}_{\mathrm{p}}^{(t)}. \tag{83}$$

To apply the Gauss–Newton update, we represent the complex channel parameter by its real-valued form:

$$\bar{\boldsymbol{\theta}}_{\mathrm{c}} = \left[\mathrm{Re}\{\boldsymbol{\theta}_{\mathrm{c}}\}^T \quad \mathrm{Im}\{\boldsymbol{\theta}_{\mathrm{c}}\}^T\right]^T \in \mathbb{R}^{2D_{\mathrm{c}}}. \tag{84}$$

The real-valued Jacobian at the t-th iteration is

$$\mathbf{J}_{\mathrm{p}}^{(t)} = 2\left[\mathrm{Re}\left\{\mathrm{diag}\left((\mathbf{u}_{\mathrm{p}}^{(t)})^*\right)\mathbf{G}_{\mathrm{p}}\right\}, -\mathrm{Im}\left\{\mathrm{diag}\left((\mathbf{u}_{\mathrm{p}}^{(t)})^*\right)\mathbf{G}_{\mathrm{p}}\right\}\right] \tag{85}$$

Then the damped Gauss–Newton increment is obtained by solving

$$\Delta\bar{\boldsymbol{\theta}}_{\mathrm{c}}^{(t)} = \left[(\mathbf{J}_{\mathrm{p}}^{(t)})^T\mathbf{J}_{\mathrm{p}}^{(t)} + \lambda^{(t)}\mathbf{I}\right]^{-1}(\mathbf{J}_{\mathrm{p}}^{(t)})^T\mathbf{e}^{(t)}, \tag{86}$$

where $\lambda^{(t)} \geq 0$ is a damping parameter used to improve numerical stability. The real-valued channel parameter is updated as

$$\bar{\boldsymbol{\theta}}_{\mathrm{c}}^{(t+1)} = \bar{\boldsymbol{\theta}}_{\mathrm{c}}^{(t)} + \Delta\bar{\boldsymbol{\theta}}_{\mathrm{c}}^{(t)}. \tag{87}$$

After convergence, the complex channel parameter $\hat{\boldsymbol{\theta}}_{\mathrm{c}}$ is reconstructed from $\bar{\boldsymbol{\theta}}_{\mathrm{c}}$.

A practical initialization can be obtained from a reference-dominated linear approximation. When the reference wave is sufficiently strong compared with the unknown field, the quadratic term $|\mathbf{G}_{\mathrm{p}}\boldsymbol{\theta}_{\mathrm{c}}|^2$ can be ignored in the early initialization stage, leading to

$$\mathbf{z}_{\mathrm{p}} - |\mathbf{b}_{\mathrm{p}}|^2 \approx 2\,\mathrm{Re}\left\{\mathbf{b}_{\mathrm{p}}^* \odot \mathbf{G}_{\mathrm{p}}\boldsymbol{\theta}_{\mathrm{c}}\right\}. \tag{88}$$

This real-valued linear model can be solved by least squares to provide an initial estimate for the iterative refinement. Alternatively, random initialization or a coarse channel prior can be used when such information is available.

The complete reference-assisted channel estimation procedure is summarized in Algorithm 2.

The complexity of each Gauss–Newton iteration is mainly determined by forming the Jacobian and solving the linear system in (86). For a channel parameter dimension D_{c} and Q_{p} training intensity observations, the per-iteration complexity is approximately

$$\mathcal{O}\left(Q_{\mathrm{p}}D_{\mathrm{c}}^2 + D_{\mathrm{c}}^3\right). \tag{89}$$

Thus, the total complexity is

$$\mathcal{O}\left(I_{\mathrm{c}}\left(Q_{\mathrm{p}}D_{\mathrm{c}}^2 + D_{\mathrm{c}}^3\right)\right), \tag{90}$$

Algorithm 2 Reference-Assisted Channel Estimation

Require: Training intensity measurements $\mathbf{z}_{\mathrm{p}}$, training matrix $\mathbf{G}_{\mathrm{p}}$, reference vector $\mathbf{b}_{\mathrm{p}}$, maximum iteration number I_{c}, and stopping threshold ϵ_{c}.

Ensure: Estimated channel parameter $\hat{\boldsymbol{\theta}}_{\mathrm{c}}$ and reconstructed channel $\hat{\mathbf{H}}$.

Initialize $\boldsymbol{\theta}_{\mathrm{c}}^{(0)}$ using (88) or a coarse prior.
for $t = 0, 1, \ldots, I_{\mathrm{c}} - 1$ **do**
  Compute
  $$\mathbf{u}_{\mathrm{p}}^{(t)} = \mathbf{G}_{\mathrm{p}}\boldsymbol{\theta}_{\mathrm{c}}^{(t)} + \mathbf{b}_{\mathrm{p}}.$$
  Compute the residual
  $$\mathbf{e}^{(t)} = \mathbf{z}_{\mathrm{p}} - |\mathbf{u}_{\mathrm{p}}^{(t)}|^2.$$
  Construct the Jacobian $\mathbf{J}_{\mathrm{p}}^{(t)}$ according to (85).
  Compute the damped Gauss–Newton increment from (86).
  Update $\bar{\boldsymbol{\theta}}_{\mathrm{c}}^{(t+1)}$ according to (87).
  if $\|\Delta\bar{\boldsymbol{\theta}}_{\mathrm{c}}^{(t)}\|_2/\|\bar{\boldsymbol{\theta}}_{\mathrm{c}}^{(t)}\|_2 < \epsilon_{\mathrm{c}}$ **then**
    break
  end if
end for
Reconstruct $\hat{\boldsymbol{\theta}}_{\mathrm{c}}$ from its real-valued representation.
Reconstruct $\hat{\mathbf{H}}$ from $\hat{\boldsymbol{\theta}}_{\mathrm{c}}$.

where I_{c} is the number of iterations. This complexity highlights the benefit of exploiting the structured far-field channel model. If the full channel vector $\mathrm{vec}(\mathbf{H})$ is estimated, then $D_{\mathrm{c}} = N_{\mathrm{r}}L$. In contrast, if the angles of arrival are known or pre-estimated and only the complex path gains are estimated, then $D_{\mathrm{c}} = L$, which significantly reduces the computational cost.

The estimated channel $\hat{\mathbf{H}}$ is then used to construct the data-phase measurement matrix $\mathbf{G}_{\mathrm{d}}(\hat{\mathbf{H}})$ for symbol detection. The impact of the channel estimation error on the subsequent data detection will be analyzed later in this section.

C. Reference-Assisted Symbol Detection

We next consider the data-symbol detection problem in the data transmission phase. After the training phase, the receiver obtains the channel estimate $\hat{\mathbf{H}}$ and constructs the data-phase equivalent measurement matrix $\mathbf{G}_{\mathrm{d}}(\hat{\mathbf{H}})$. For notational simplicity, we denote it by $\hat{\mathbf{G}}_{\mathrm{d}}$ in this subsection. The data-phase intensity observation is given by

$$\mathbf{z}_{\mathrm{d}} = \left|\hat{\mathbf{G}}_{\mathrm{d}}\mathbf{s}_{\mathrm{d}} + \mathbf{b}_{\mathrm{d}}\right|^2 + \mathbf{w}_{\mathrm{d}}, \tag{91}$$

where $\mathbf{s}_{\mathrm{d}} \in \mathcal{S}^{D_{\mathrm{s}}}$ is the unknown data-symbol vector, $D_{\mathrm{s}} = LK_{\mathrm{d}}$ is the total number of data symbols in the considered block, $\mathbf{b}_{\mathrm{d}}$ is the designed data-phase reference vector, and $\mathbf{w}_{\mathrm{d}}$ is the effective post-detection noise.

1) Maximum-Likelihood Detection: When the post-detection noise is modeled as additive white Gaussian noise in the intensity domain, the maximum-likelihood detector is equivalent to the following least-squares detector:

$$\hat{\mathbf{s}}_{\mathrm{ML}} = \arg\min_{\mathbf{s}\in\mathcal{S}^{D_{\mathrm{s}}}} \left\| \mathbf{z}_{\mathrm{d}} - \left|\hat{\mathbf{G}}_{\mathrm{d}}\mathbf{s} + \mathbf{b}_{\mathrm{d}}\right|^2 \right\|_2^2. \tag{92}$$

The ML detector directly searches over the finite constellation set and therefore provides a natural performance benchmark for reference-assisted symbol detection. Its reliability is closely related to the energy-domain minimum distance defined in Section III. Specifically, if the designed reference vector enlarges

$$d_{\min,\mathrm{d}}^{(E)} = \min_{\mathbf{s}_i\neq\mathbf{s}_j} \left\| \left|\hat{\mathbf{G}}_{\mathrm{d}}\mathbf{s}_i + \mathbf{b}_{\mathrm{d}}\right|^2 - \left|\hat{\mathbf{G}}_{\mathrm{d}}\mathbf{s}_j + \mathbf{b}_{\mathrm{d}}\right|^2 \right\|_2, \tag{93}$$

then the pairwise separation between candidate symbols is improved, leading to a lower probability of symbol confusion.

However, the complexity of exhaustive ML detection grows exponentially with the number of detected symbols. For a constellation size $|\mathcal{S}|$, the search space has cardinality $|\mathcal{S}|^{D_{\mathrm{s}}}$. The corresponding complexity is approximately

$$\mathcal{O}\left(|\mathcal{S}|^{D_{\mathrm{s}}} Q_{\mathrm{d}}\right), \tag{94}$$

where Q_{d} is the number of data-phase intensity observations. Therefore, ML detection is mainly suitable for short symbol blocks or for benchmarking lower-complexity detectors.

2) Projected Gauss–Newton Detection: To reduce the detection complexity, we also consider a projected Gauss–Newton detector. The main idea is to first relax the finite-alphabet constraint and solve a continuous nonlinear least-squares problem:

$$\hat{\mathbf{s}}_{\mathrm{cont}} = \arg\min_{\mathbf{s}\in\mathbb{C}^{D_{\mathrm{s}}}} \left\| \mathbf{z}_{\mathrm{d}} - \left|\hat{\mathbf{G}}_{\mathrm{d}}\mathbf{s} + \mathbf{b}_{\mathrm{d}}\right|^2 \right\|_2^2, \tag{95}$$

and then project the obtained continuous estimate onto the constellation set.

Let

$$\mathbf{u}_{\mathrm{d}}^{(t)} = \hat{\mathbf{G}}_{\mathrm{d}}\mathbf{s}^{(t)} + \mathbf{b}_{\mathrm{d}} \tag{96}$$

be the predicted complex field at the t-th iteration. The predicted intensity and residual are

$$\hat{\mathbf{z}}_{\mathrm{d}}^{(t)} = |\mathbf{u}_{\mathrm{d}}^{(t)}|^2, \tag{97}$$

and

$$\mathbf{e}_{\mathrm{d}}^{(t)} = \mathbf{z}_{\mathrm{d}} - \hat{\mathbf{z}}_{\mathrm{d}}^{(t)}. \tag{98}$$

Similar to the channel estimation stage, define the real-valued symbol vector as

$$\bar{\mathbf{s}} = \left[\mathrm{Re}\{\mathbf{s}\}^T \quad \mathrm{Im}\{\mathbf{s}\}^T\right]^T \in \mathbb{R}^{2D_{\mathrm{s}}}. \tag{99}$$

The corresponding real-valued Jacobian is

$$\mathbf{J}_{\mathrm{d}}^{(t)} = 2\Big[\mathrm{Re}\Big\{\mathrm{diag}\big((\mathbf{u}_{\mathrm{d}}^{(t)})^*\big)\hat{\mathbf{G}}_{\mathrm{d}}\Big\},\ -\mathrm{Im}\Big\{\mathrm{diag}\big((\mathbf{u}_{\mathrm{d}}^{(t)})^*\big)\hat{\mathbf{G}}_{\mathrm{d}}\Big\}\Big]. \tag{100}$$

The damped Gauss–Newton increment is computed as

$$\Delta\bar{\mathbf{s}}^{(t)} = \left[(\mathbf{J}_{\mathrm{d}}^{(t)})^T\mathbf{J}_{\mathrm{d}}^{(t)} + \mu^{(t)}\mathbf{I}\right]^{-1}(\mathbf{J}_{\mathrm{d}}^{(t)})^T\mathbf{e}_{\mathrm{d}}^{(t)}, \tag{101}$$

where $\mu^{(t)} \geq 0$ is a damping factor. The continuous estimate is updated by

$$\bar{\mathbf{s}}^{(t+1)} = \bar{\mathbf{s}}^{(t)} + \Delta\bar{\mathbf{s}}^{(t)}. \tag{102}$$

Algorithm 3 Projected Gauss–Newton Symbol Detection

Require: Data intensity measurements $\mathbf{z}_{\mathrm{d}}$, equivalent matrix $\hat{\mathbf{G}}_{\mathrm{d}}$, reference vector $\mathbf{b}_{\mathrm{d}}$, constellation set $\mathcal{S}$, maximum iteration number I_{d}, and stopping threshold ϵ_{d}.

Ensure: Detected symbol vector $\hat{\mathbf{s}}_{\mathrm{d}}$.

Initialize $\mathbf{s}^{(0)}$ randomly, from a linearized estimate, or from the previous detected block.
for $t = 0, 1, \ldots, I_{\mathrm{d}} - 1$ **do**
  Compute
$$\mathbf{u}_{\mathrm{d}}^{(t)} = \hat{\mathbf{G}}_{\mathrm{d}}\mathbf{s}^{(t)} + \mathbf{b}_{\mathrm{d}}.$$
  Compute the residual
$$\mathbf{e}_{\mathrm{d}}^{(t)} = \mathbf{z}_{\mathrm{d}} - |\mathbf{u}_{\mathrm{d}}^{(t)}|^2.$$
  Construct the Jacobian $\mathbf{J}_{\mathrm{d}}^{(t)}$ according to (100).
  Compute the damped Gauss–Newton increment from (101).
  Update the continuous estimate according to (102).
  Optionally project $\mathbf{s}^{(t+1)}$ onto $\mathcal{S}^{D_{\mathrm{s}}}$ using (103).
  if $\|\Delta\bar{\mathbf{s}}^{(t)}\|_2/\|\bar{\mathbf{s}}^{(t)}\|_2 < \epsilon_{\mathrm{d}}$ **then**
    break
  end if
end for
Output
$$\hat{\mathbf{s}}_{\mathrm{d}} = \mathcal{Q}_{\mathcal{S}}\left(\mathbf{s}^{(t+1)}\right).$$

After each iteration, or after convergence, the continuous estimate can be projected onto the constellation set:

$$\mathbf{s}^{(t+1)} \leftarrow \mathcal{Q}_{\mathcal{S}}\left(\mathbf{s}^{(t+1)}\right), \tag{103}$$

where $\mathcal{Q}_{\mathcal{S}}(\cdot)$ denotes element-wise nearest-neighbor quantization onto $\mathcal{S}$. If the projection is applied after each iteration, the algorithm becomes a projected Gauss–Newton detector. If the projection is applied only after convergence, the algorithm first performs continuous phase retrieval and then makes a final constellation decision.

The projected Gauss–Newton detector is summarized in Algorithm 3.

3) Biased Gerchberg–Saxton Detection: As another low-complexity alternative, a biased Gerchberg–Saxton detector can be used by exploiting the known reference vector. At the t-th iteration, the current field estimate is

$$\mathbf{u}_{\mathrm{d}}^{(t)} = \hat{\mathbf{G}}_{\mathrm{d}}\mathbf{s}^{(t)} + \mathbf{b}_{\mathrm{d}}. \tag{104}$$

The magnitude constraint imposed by the intensity measurements is enforced as

$$\tilde{\mathbf{u}}_{\mathrm{d}}^{(t)} = \sqrt{\mathbf{z}_{\mathrm{d}}} \odot e^{j\angle\mathbf{u}_{\mathrm{d}}^{(t)}}. \tag{105}$$

Then the unknown symbol vector is updated by removing the reference vector and projecting back through the measurement matrix:

$$\mathbf{s}^{(t+1)} = \hat{\mathbf{G}}_{\mathrm{d}}^{\dagger}\left(\tilde{\mathbf{u}}_{\mathrm{d}}^{(t)} - \mathbf{b}_{\mathrm{d}}\right), \tag{106}$$

where $\hat{\mathbf{G}}_\mathrm{d}^\dagger$ denotes the Moore–Penrose pseudo-inverse of $\hat{\mathbf{G}}_\mathrm{d}$. Finally, the updated estimate is projected onto the constellation set:

$$\mathbf{s}^{(t+1)} \leftarrow \mathcal{Q}_\mathcal{S}\left(\mathbf{s}^{(t+1)}\right). \tag{107}$$

This method is simple and computationally efficient when the pseudo-inverse can be precomputed. However, its performance depends on the conditioning of $\hat{\mathbf{G}}_\mathrm{d}$ and on the quality of the reference-assisted measurements.

4) Complexity Discussion: The complexity of ML detection is exponential in the number of detected symbols, as shown in (94). In contrast, the projected Gauss–Newton detector has per-iteration complexity

$$\mathcal{O}\left(Q_\mathrm{d} D_\mathrm{s}^2 + D_\mathrm{s}^3\right), \tag{108}$$

which comes from forming the Jacobian and solving the damped normal equation. The biased Gerchberg–Saxton detector has lower per-iteration complexity. If $\hat{\mathbf{G}}_\mathrm{d}^\dagger$ is precomputed, each iteration mainly requires matrix-vector multiplications, with complexity approximately

$$\mathcal{O}\left(Q_\mathrm{d} D_\mathrm{s}\right). \tag{109}$$

Therefore, ML detection is suitable as a benchmark for small-scale systems, while projected Gauss–Newton and biased Gerchberg–Saxton detectors provide scalable alternatives for larger data blocks.

The choice of detector also depends on the reference-wave design. A larger energy-domain minimum distance improves the reliability of ML detection and provides a more favorable objective landscape for iterative detectors. Thus, the reference design in Section IV and the detection algorithms in this section are complementary: the former improves the separability of the intensity-domain observations, while the latter recovers the transmitted symbols from these observations.

D. Impact of Channel Estimation Error and Complexity Analysis

The preceding symbol detection algorithms assume that the data-phase equivalent measurement matrix is constructed from the estimated channel. Therefore, the channel estimation error inevitably introduces model mismatch in the data detection stage. In this subsection, we analyze the impact of this mismatch and summarize the computational complexity of the proposed two-phase recovery procedure.

Let the estimated channel be expressed as

$$\hat{\mathbf{H}} = \mathbf{H} + \Delta\mathbf{H}, \tag{110}$$

where $\Delta\mathbf{H}$ denotes the channel estimation error. Accordingly, the data-phase equivalent measurement matrix can be written as

$$\hat{\mathbf{G}}_\mathrm{d} = \mathbf{G}_\mathrm{d} + \Delta\mathbf{G}_\mathrm{d}, \tag{111}$$

where $\mathbf{G}_\mathrm{d}$ is the ideal measurement matrix constructed from the true channel, and $\Delta\mathbf{G}_\mathrm{d}$ is the corresponding perturbation induced by $\Delta\mathbf{H}$.

The true data-phase intensity observation is

$$\mathbf{z}_\mathrm{d} = \left|\mathbf{G}_\mathrm{d}\mathbf{s}_\mathrm{d} + \mathbf{b}_\mathrm{d}\right|^2 + \mathbf{w}_\mathrm{d}. \tag{112}$$

However, the detector evaluates candidate symbols using the mismatched model

$$\hat{\mathcal{T}}_\mathrm{d}(\mathbf{s}) = \left|\hat{\mathbf{G}}_\mathrm{d}\mathbf{s} + \mathbf{b}_\mathrm{d}\right|^2. \tag{113}$$

For a given candidate symbol vector $\mathbf{s}$, the intensity-domain mismatch caused by channel estimation error is

$$\Delta\mathbf{z}_\mathrm{CSI}(\mathbf{s}) = \left|(\mathbf{G}_\mathrm{d} + \Delta\mathbf{G}_\mathrm{d})\mathbf{s} + \mathbf{b}_\mathrm{d}\right|^2 - \left|\mathbf{G}_\mathrm{d}\mathbf{s} + \mathbf{b}_\mathrm{d}\right|^2. \tag{114}$$

When $\Delta\mathbf{G}_\mathrm{d}$ is small, the first-order approximation gives

$$\Delta\mathbf{z}_\mathrm{CSI}(\mathbf{s}) \approx 2\,\mathrm{Re}\left\{(\mathbf{G}_\mathrm{d}\mathbf{s} + \mathbf{b}_\mathrm{d})^* \odot \Delta\mathbf{G}_\mathrm{d}\mathbf{s}\right\}. \tag{115}$$

This expression shows that channel estimation error acts as an additional intensity-domain perturbation. Its impact depends not only on the channel error magnitude, but also on the reference-assisted field $\mathbf{G}_\mathrm{d}\mathbf{s} + \mathbf{b}_\mathrm{d}$ and the symbol-dependent perturbation $\Delta\mathbf{G}_\mathrm{d}\mathbf{s}$.

The robustness of data detection under channel estimation error can be interpreted through the energy-domain distance. For two candidate symbol vectors $\mathbf{s}_i$ and $\mathbf{s}_j$, define the ideal pairwise intensity separation as

$$d_{ij}^{(E)} = \left\|\left|\mathbf{G}_\mathrm{d}\mathbf{s}_i + \mathbf{b}_\mathrm{d}\right|^2 - \left|\mathbf{G}_\mathrm{d}\mathbf{s}_j + \mathbf{b}_\mathrm{d}\right|^2\right\|_2. \tag{116}$$

Due to post-detection noise and channel estimation error, the effective perturbation in the data detection stage can be approximated as

$$\mathbf{e}_\mathrm{eff} = \mathbf{w}_\mathrm{d} + \Delta\mathbf{z}_\mathrm{CSI}(\mathbf{s}_\mathrm{d}). \tag{117}$$

A sufficient condition for correctly distinguishing $\mathbf{s}_\mathrm{d}$ from another candidate $\mathbf{s}'$ is

$$d_E(\mathbf{s}_\mathrm{d}, \mathbf{s}') > 2\|\mathbf{e}_\mathrm{eff}\|_2. \tag{118}$$

Therefore, a larger energy-domain minimum distance provides higher tolerance to both post-detection noise and channel estimation error. This observation further justifies the reference design criterion in Section IV, which aims to enlarge $d_\mathrm{min,d}^{(E)}$ for finite-alphabet symbol detection.

The channel estimation quality can be evaluated by the normalized mean-square error (NMSE)

$$\mathrm{NMSE}_\mathrm{H} = \frac{\mathbb{E}\left\{\|\hat{\mathbf{H}} - \mathbf{H}\|_F^2\right\}}{\mathbb{E}\left\{\|\mathbf{H}\|_F^2\right\}}. \tag{119}$$

In the simulations, we will compare the data detection performance under perfect CSI and estimated CSI to quantify the effect of channel estimation error on the end-to-end system performance.

Finally, we summarize the computational complexity of the main estimation and detection procedures. For reference-assisted channel estimation using damped Gauss–Newton iterations, the per-iteration complexity is dominated by forming the Jacobian and solving the damped normal equation. For a channel parameter dimension D_c and Q_p training observations, the total complexity is approximately

$$\mathcal{O}\left(I_\mathrm{c}\left(Q_\mathrm{p} D_\mathrm{c}^2 + D_\mathrm{c}^3\right)\right), \tag{120}$$

where I_c is the number of channel-estimation iterations. If the full channel is estimated, then $D_\mathrm{c} = N_\mathrm{r} L$. If the far-field structure is exploited and only the complex path gains are estimated, then $D_\mathrm{c} = L$, resulting in a much lower computational cost.

For data detection, exhaustive ML detection has complexity

$$\mathcal{O}\left(|\mathcal{S}|^{D_\mathrm{s}} Q_\mathrm{d}\right), \tag{121}$$

where $D_\mathrm{s} = LK_\mathrm{d}$ is the number of detected data symbols and Q_d is the number of data-phase intensity observations. The projected Gauss–Newton detector has total complexity

$$\mathcal{O}\left(I_\mathrm{d}\left(Q_\mathrm{d} D_\mathrm{s}^2 + D_\mathrm{s}^3\right)\right), \tag{122}$$

where I_d is the number of detection iterations. The biased Gerchberg–Saxton detector has lower complexity. When the pseudo-inverse of $\hat{\mathbf{G}}_\mathrm{d}$ is precomputed, each iteration mainly involves matrix-vector multiplications, leading to

$$\mathcal{O}\left(I_\mathrm{d} Q_\mathrm{d} D_\mathrm{s}\right). \tag{123}$$

The above analysis shows that the proposed two-phase receiver can trade off detection performance and computational complexity. ML detection provides a benchmark for small-scale symbol blocks, while projected Gauss–Newton and biased Gerchberg–Saxton detectors provide scalable alternatives. Moreover, exploiting the structured far-field channel model significantly reduces the channel-estimation dimension. Together with the reference-wave design in Section IV, these algorithms provide a practical implementation of the proposed reference-assisted holographic RIS receiver.

V. Simulation Results

In this section, numerical simulations are conducted to evaluate the proposed reference-assisted holographic RIS receiver. We first examine the channel estimation performance in the training phase, then investigate the symbol detection performance under both perfect and estimated channel state information (CSI). Finally, we study the impact of reference-wave design and RIS measurement diversity on the energy-domain separability and detection reliability.

A. Simulation Setup

We consider an uplink multi-user communication system assisted by a holographic RIS receiver. Unless otherwise specified, the RIS receiver is modeled as a uniform planar array placed on the yz-plane with $N_y \times N_z$ receiving elements. The total number of RIS elements is

$$N_\mathrm{r} = N_y N_z. \tag{124}$$

The inter-element spacings are set as

$$d_y = d_z = \frac{\lambda}{2}, \tag{125}$$

where λ is the carrier wavelength. The carrier frequency is denoted by f_c, and the corresponding wavelength is $\lambda = c/f_c$, where c is the speed of light.

The far-field channel model in Section II is adopted. The channel matrix is generated as

$$\mathbf{H} = \mathbf{V}\boldsymbol{\Gamma}, \tag{126}$$

where $\mathbf{V}$ is the RIS array manifold matrix and $\boldsymbol{\Gamma} = \mathrm{diag}(\alpha_1, \ldots, \alpha_L)$ contains the complex channel gains of the L users. The channel gains are modeled as

$$\alpha_l = \sqrt{\beta_l} e^{j\omega_l}, \tag{127}$$

where β_l denotes the large-scale channel power gain and ω_l is uniformly distributed over $[0, 2\pi)$. The users' angles of arrival are selected according to the considered simulation scenario. In particular, to study the effect of spatial correlation, the angular separation between users will be varied in some simulations.

For the transmitter, each user transmits symbols drawn from a normalized constellation $\mathcal{S}$, such as QPSK or 16-QAM. The pulse-shaping matrix is constructed according to the pulse-shaping filter introduced in Section II. Unless otherwise specified, the data block contains K_d symbols per user, and the total data-symbol dimension is

$$D_\mathrm{s} = LK_\mathrm{d}. \tag{128}$$

During the training phase, each user transmits known pilot symbols over T_p pilot observations.

The received intensity measurements are generated according to the reference-assisted square-law model

$$\mathbf{z} = |\mathbf{G}\boldsymbol{\theta} + \mathbf{b}|^2 + \mathbf{w}, \tag{129}$$

where $\mathbf{w}$ is modeled as additive real-valued Gaussian noise in the intensity domain. The signal-to-noise ratio (SNR) is defined as

$$\mathrm{SNR} = 10\log_{10}\left(\frac{\mathbb{E}\left\{\left\||\mathbf{G}\boldsymbol{\theta} + \mathbf{b}|^2\right\|_2^2\right\}}{\mathbb{E}\{\|\mathbf{w}\|_2^2\}}\right). \tag{130}$$

We compare the following reference-wave designs:

- **No reference**: $\mathbf{b} = \mathbf{0}$.
- **Constant reference**: all reference entries have the same phase.
- **Random reference**: the reference phases are independently drawn from $\mathcal{U}[0, 2\pi)$.
- **Uniform reference**: the reference phases are uniformly distributed over $[0, 2\pi)$.
- **Codebook-optimized reference**: the reference pattern is selected from a finite codebook according to the design criteria in Section IV.

For fair comparison, the total reference power is kept identical among different reference designs unless otherwise stated.

For channel estimation, the damped Gauss–Newton algorithm in Section V is used. The estimation performance is measured by the normalized mean-square error (NMSE)

$$\mathrm{NMSE}_\mathrm{H} = \frac{\mathbb{E}\left\{\|\hat{\mathbf{H}} - \mathbf{H}\|_F^2\right\}}{\mathbb{E}\left\{\|\mathbf{H}\|_F^2\right\}}. \tag{131}$$

For symbol detection, we compare maximum-likelihood (ML) detection, projected Gauss–Newton detection, and biased Gerchberg–Saxton (GS) detection. The detection performance is evaluated using the symbol error rate (SER), defined as

$$\mathrm{SER} = \frac{\text{number of incorrectly detected symbols}}{\text{total number of transmitted symbols}}. \tag{132}$$

TABLE I: Default Simulation Parameters

Parameter	Default value
Carrier frequency f_c	3.5 GHz
RIS array size	$N_y \times N_z = 8 \times 8$
Number of RIS elements N_{r}	64
Element spacing	$d_y = d_z = \lambda/2$
Number of users L	2 or 4
Modulation constellation $\mathcal{S}$	QPSK / 16-QAM
Pilot length T_{p}	scenario-dependent
Data block length K_{d}	scenario-dependent
Reference amplitude	fixed unless otherwise stated
Reference phase designs	none / constant / random / uniform / codebook
Channel estimator	damped Gauss–Newton
Symbol detectors	ML / projected GN / biased GS
Performance metrics	NMSE / SER / $d_{\min}^{(E)}$

To connect the simulations with the theoretical analysis, we also evaluate the energy-domain minimum distance

$$d_{\min}^{(E)} = \min_{\mathbf{s}_i \neq \mathbf{s}_j} \left\| |\mathbf{G}_{\mathrm{d}}\mathbf{s}_i + \mathbf{b}_{\mathrm{d}}|^2 - |\mathbf{G}_{\mathrm{d}}\mathbf{s}_j + \mathbf{b}_{\mathrm{d}}|^2 \right\|_2, \tag{133}$$

as well as the singular-value and condition-number metrics associated with the effective measurement matrix and the Jacobian. All results are averaged over independent channel realizations, noise realizations, and transmitted data blocks.

The default simulation parameters are summarized in Table I. Unless otherwise stated, these default values are used in the following simulations.

B. Channel Estimation Performance

In this subsection, we evaluate the performance of the proposed reference-assisted channel estimation scheme in the training phase. The main objective is to verify the role of the reference wave in improving the local identifiability and stability of channel estimation. Unless otherwise specified, the default simulation parameters in Table I are adopted.

We first compare the channel estimation accuracy under different reference-wave designs. The NMSE performance versus SNR is shown in Fig. **??**. The compared schemes include no reference, constant-phase reference, random-phase reference, uniform-phase reference, and codebook-optimized reference. This comparison is used to verify whether reference phase diversity can improve the conditioning of the intensity-based channel estimation problem.

Next, we study the effect of the pilot length on channel estimation. According to the local identifiability analysis, the number of intensity observations should be sufficiently large to make the Jacobian full column rank and well conditioned. Fig. **??** shows the NMSE as a function of the pilot length T_{p}. This result is intended to demonstrate how additional pilot observations improve the stability of reference-assisted channel estimation.

We then evaluate the influence of reference power. The reference wave provides phase anchoring through the interference term, but its amplitude also affects the dynamic range and the sensitivity of the square-law measurements. Fig. **??** presents the NMSE performance under different reference amplitudes. This figure is used to determine a reasonable reference-power region for the proposed receiver.

To connect the simulation results with the Jacobian-based analysis, we also evaluate the singular-value behavior of the real-valued Jacobian. Fig. **??** shows the minimum singular value and/or condition number of the Jacobian under different reference-wave designs. This result is used to verify that reference patterns leading to larger $\sigma_{\min}(\mathbf{J}_{\mathrm{p}})$ or smaller $\kappa(\mathbf{J}_{\mathrm{p}})$ also lead to improved NMSE performance.

Finally, we examine the effect of user angular separation. For the far-field channel model, the RIS array manifold matrix $\mathbf{V}$ becomes ill-conditioned when the users have close angles of arrival. This reduces the separability of the channel parameters and degrades the channel estimation performance. Fig. **??** shows the NMSE versus the angular separation between users. The corresponding condition number of the array manifold matrix can also be plotted to illustrate the relationship between spatial correlation and estimation accuracy.

Overall, these results are expected to validate the theoretical insights in Sections III and IV: the training reference wave improves channel estimation by enhancing the local sensitivity of the intensity mapping, while the RIS array manifold and pilot structure determine whether the channel parameters are sufficiently distinguishable in the complex field domain.

C. Symbol Detection Performance with Perfect CSI

In this subsection, we evaluate the data-symbol detection performance under perfect CSI. The purpose is to isolate the impact of reference-assisted intensity measurements and detection algorithms from channel estimation errors. Therefore, the data-phase equivalent measurement matrix $\mathbf{G}_{\mathrm{d}}$ is constructed using the true channel matrix $\mathbf{H}$. Unless otherwise specified, the default simulation parameters in Table I are adopted.

We first compare different symbol detection algorithms under the same reference-wave design. Fig. **??** shows the SER versus SNR for maximum-likelihood (ML) detection, projected Gauss–Newton (GN) detection, and biased Gerchberg–Saxton (GS) detection. The ML detector serves as a benchmark, while projected GN and biased GS provide lower-complexity alternatives. This comparison evaluates the performance-complexity tradeoff of the proposed reference-assisted detection framework.

Next, we investigate the impact of reference-wave design on symbol detection. Fig. **??** compares the SER performance of different reference structures, including no reference, constant-phase reference, random-phase reference, uniform-phase reference, and codebook-optimized reference. For a fair comparison, the total reference power is kept the same for all reference-assisted schemes. This figure is intended to verify that reference phase diversity improves the separability of finite-alphabet symbols in the intensity domain.

To directly validate the energy-domain minimum distance analysis, Fig. **??** presents $d_{\min}^{(E)}$ under different reference-wave designs. According to the theoretical analysis, a larger $d_{\min}^{(E)}$ implies stronger pairwise separability after square-law detection. This result is used to explain the SER trends observed in Fig. **??**.

We further examine the relationship between the energy-domain minimum distance and the detection performance.

Fig. **??** plots the SER as a function of $d_{\min}^{(E)}$ by varying the reference pattern or reference phase codebook. This figure is used to verify that improving the energy-domain minimum distance leads to a lower symbol error rate, which supports the max-min reference design criterion introduced in Section IV.

Finally, we study the effect of modulation order on the proposed detection framework. Fig. **??** compares the SER performance for different constellations, such as QPSK and 16-QAM. Higher-order modulation leads to denser constellation points and smaller effective pairwise separations, making symbol detection more sensitive to intensity-domain ambiguity and post-detection noise.

Overall, the perfect-CSI results are expected to demonstrate that the proposed reference-assisted receiver can reliably recover finite-alphabet symbols from intensity-only measurements when the reference wave is properly designed. They also show that the energy-domain minimum distance provides a meaningful indicator for predicting and improving symbol detection performance.

D. End-to-End Symbol Detection with Estimated CSI

In this subsection, we evaluate the end-to-end data detection performance when the channel is estimated from the training phase. Different from the perfect-CSI results in the previous subsection, the data-phase equivalent measurement matrix is now constructed from the estimated channel $\hat{\mathbf{H}}$. Therefore, the detection performance is affected by both post-detection noise and channel estimation error. Unless otherwise specified, the same reference design is used consistently in the training and data phases, and the default simulation parameters in Table I are adopted.

We first compare the SER performance under perfect CSI and estimated CSI. Fig. **??** shows the SER versus SNR for the proposed receiver using different symbol detectors. The perfect-CSI curves provide performance benchmarks, while the estimated-CSI curves quantify the degradation caused by channel estimation error. This comparison validates the necessity of accurate reference-assisted channel estimation for reliable data detection.

Next, we investigate the impact of pilot length on the end-to-end detection performance. Fig. **??** shows the SER as a function of the pilot length T_{p}. A larger pilot length generally improves the channel estimate, which in turn reduces the mismatch in the data-phase measurement matrix. This result links the channel estimation accuracy in the training phase to the final symbol detection reliability.

To further quantify the influence of channel estimation error, Fig. **??** plots the SER as a function of the channel estimation NMSE. This can be obtained either by using different pilot lengths and training SNRs or by injecting controlled channel perturbations into the true channel. The purpose of this figure is to verify that a larger channel estimation error leads to a stronger intensity-domain model mismatch and thus degrades symbol detection performance.

We also compare the end-to-end performance under different training reference designs. Fig. **??** shows the SER obtained with no reference, constant-phase reference, random-phase reference, uniform-phase reference, and codebook-optimized reference in the training phase. This comparison demonstrates how the training reference design affects the final data detection performance through the quality of the estimated channel.

Finally, we evaluate the impact of using different reference designs in the data phase under estimated CSI. Fig. **??** compares the SER performance of different data-phase references while using the same channel estimation procedure. This result separates the effect of data-phase symbol separability from the effect of training-phase channel estimation accuracy.

Overall, these results are expected to show that the proposed reference-assisted receiver remains effective under estimated CSI, but its performance depends on both the training-stage channel estimation quality and the data-stage energy-domain separability. They also verify the analysis in Section V: channel estimation error introduces additional intensity-domain mismatch, while a larger energy-domain minimum distance provides better robustness to this mismatch.

E. Effect of Reference and RIS Measurement Diversity

In this subsection, we further investigate how reference diversity and RIS measurement diversity affect the performance of the proposed receiver. The purpose is to verify the design principle developed in Section IV: the effective measurement matrix should preserve the relevant complex-field differences, while the reference wave should project these differences into distinguishable intensity-domain observations. Unless otherwise specified, the default simulation parameters in Table I are adopted.

We first study the impact of the number of RIS receiving elements. Fig. **??** shows the energy-domain minimum distance $d_{\min}^{(E)}$ as a function of the number of RIS elements N_{r}. As the number of RIS elements increases, the receiver obtains more spatial intensity observations, which generally improves the separability of different symbol vectors in the energy domain.

The corresponding SER performance is shown in Fig. **??**. This figure is used to verify that the improvement in $d_{\min}^{(E)}$ obtained from additional RIS observations can be translated into lower symbol detection error rates. Together with Fig. **??**, this result highlights the role of the holographic RIS as a spatial measurement device rather than merely a passive aperture.

Next, we evaluate the effect of multiple reference states. Fig. **??** plots $d_{\min}^{(E)}$ versus the number of reference states R. Each reference state provides an additional intensity projection of the same underlying complex field. Therefore, increasing R improves measurement diversity and can reduce the probability that a vulnerable symbol pair remains weakly separated under all reference states.

Fig. **??** further shows the SER performance under different numbers of reference states. This result is intended to demonstrate the tradeoff between detection reliability and measurement overhead. Although more reference states provide stronger intensity-domain diversity, they also require additional measurement time or reference switching operations.

We then examine the effect of user angular separation. For the far-field RIS channel model, smaller angular separation leads to more correlated steering vectors and a more ill-conditioned array manifold matrix $\mathbf{V}$. Fig. **??** shows the condition number or minimum singular value of $\mathbf{V}$ as a function of the user angular separation. This figure directly illustrates how the spatial geometry affects the conditioning of the RIS measurement matrix.

The corresponding end-to-end detection performance is shown in Fig. **??**. When the users have close angles of arrival, the steering vectors become highly correlated, which reduces the effective field-domain separation and degrades both channel estimation and symbol detection. As the angular separation increases, the RIS array manifold becomes better conditioned, leading to improved detection reliability.

Finally, we compare reference-only diversity and joint reference-RIS measurement diversity. In the reference-only case, the effective measurement matrix is fixed while the reference phases are varied. In the joint diversity case, both the reference vector and the RIS measurement configuration are varied across measurement states. Fig. **??** compares the resulting $d_{\min}^{(E)}$ and/or SER performance. This comparison is used to verify that varying both $\mathbf{G}$ and $\mathbf{b}$ provides richer observations than changing the reference vector alone.

Overall, these results are expected to confirm that the proposed receiver benefits from both reference diversity and RIS spatial measurement diversity. The reference wave improves the intensity-domain projection of complex-field differences, while the RIS array and its measurement configurations determine whether these differences are sufficiently preserved before square-law detection. This validates the central design principle of the proposed reference-assisted holographic RIS receiver.

VI. Conclusion

This paper presented a comprehensive framework for signal recovery in holographic RIS receivers using only amplitude measurements. By employing a known reference wave and formulating the recovery as a nonlinear least-squares problem, we enable the estimation of both amplitude and phase of the transmitted signal. The proposed Gauss-Newton algorithm efficiently solves this optimization, with simulations demonstrating accurate recovery under various SNR conditions.

The mathematical derivation provides complete details from signal transmission through the channel, interference with the reference wave at the RIS, intensity measurement, to the final recovery algorithm. This framework serves as a foundation for more advanced RIS receiver designs, including those with adaptive reflection coefficients, multiple users, and frequency-selective channels.

Future work will extend this approach to practical implementations with hardware impairments, investigate deep learning-based recovery for reduced complexity, and explore applications in joint communication and sensing with RIS.

Appendix A

Appendix B

References

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